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Towards a definition of functional ecology

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Introduction

Organisms are open systems that exist some distance from thermodynamic equilibrium by virtue of being open to inputs and outputs of energy. Some take in energy as light, and some as organic materials from plants, other animals or even dead organisms. These processes contribute to the maintenance of the 'milieu intérieur' and fit organisms for life within particular ecological conditions.

In these opening sentences we have three senses in which ecologists might use the term 'function'. The first, following a tradition established by the Atomists, emphasizes process rather than just property. The second, following an Eltonian approach to communities, emphasizes the trophic interdependence of the parts and the 'rôle' they play within the economy of the whole. The third, following Darwin, considers organisms as outcomes of an evolutionary process driven by natural selection. All share a commonality of approach in that they focus on organisms as dynamic entities that interact with their environment as resource-processing systems. Even in this minimal sense, therefore, it might be argued that functional ecology is a relevant and appropriate way of bringing together ideas, results and techniques of ecologists working on similar themes. But here I consider if there is more to it than that; if, that is, there is an even more coherent and unitary approach that marks out functional ecology as a distinct sub-discipline. First, though, I go back to a formal consideration of the definition of 'function' and the problems it poses for biology and ecology.

Function, design and teleology

Etymologically the term 'function' is harmless enough, 'the action of performing' (Oxford English Dictionary), but it ranges in meaning from 'action in general' to 'the mode of action by which a thing

or process fulfills its purpose'. It is in this latter sense that 'function' runs into problems in biology (Woodfield, 1976) for it is then burdened with connections with the Argument from Design (whose purpose?) and teleological thinking (the effect, purpose, explains the action rather than the other way round). Darwinism does help, for 'purpose' can be construed as a contribution to the survivorship and fecundity of the organism with the trait, and the process is teleological only in the sense that the trait has been successful in the past and will be again *provided* that past environmental conditions repeat themselves. 'Function' thus becomes nearly synonymous with 'adaptation' as this term is commonly used. Yet there are still problems. For example, how do we know that the function we are ascribing to a trait is a result of natural selection and not just a figment of a vivid imagination (what Haldane called the Panglossian problem) and to what biological unit should the benefit accruing from an action be ascribed — gene, organ, organism, or even group? These are long-standing issues in biology that have proved as elusive as the philosophical puzzles behind them. There can be no question, therefore, of having the last word on them here; rather the aim is simply to demonstrate their relevance for functional ecology and to suggest ways of approaching the challenges they present.

Function and fitness

There is an immediate sense in which functions can be judged, e.g. the efficiency of photosynthesis in converting photons and raw materials to biomass, the efficiency of respiratory processes in yielding ATP, the effectiveness of leaf shape in maximizing the interception of light and minimizing water loss through transpiration, the effectiveness of respiratory structures in maximizing O₂ uptake from the ambient medium. That is, processes and structures can be assessed in terms of the ways they contribute to the economy and homeostasis of the whole, organismic system of which they are part.

But there is also a more ultimate sense in which functions can be judged; relative, that is, to their

impact on survivorship (s), fecundity (n) and developmental rates (t) — the components of Darwinian fitness. Those functions that enhance fitness (maximize s and n and minimize t) will increase in frequency in populations and it is in this sense that we can say that variants of a function are more or less well adapted. Questions can be raised about the potential circularity of this adaptationist programme; by definition, only those traits that have spread in the past (maximized s , n and minimized t) will be represented in present populations — so what? But the challenge for adaptationists is surely to discover why one variant of a trait is better (in terms of s , n , t) than another in particular ecological circumstances (Dawkins, 1982). This involves making links between the immediate and more ultimate assessments of physiological traits, and these are not always self-evident and straightforward. Moreover, it is important not to fall into the Panglossian trap noted above; not all characters will be perfect adaptations, for evolution might lag behind environmental change, or the possible extent of evolutionary change might be constrained in various ways, or a particular trait might not be an adaptation at all but simply a side-effect of others — and it is part of the responsibility of the adaptationists to distinguish between these possibilities. Much has been written about imperfection as a problem for adaptationism (e.g. Gould & Lewontin, 1979), yet perfect adaptation would probably be even more of a problem, for then it would be difficult to refute the Argument from Design; unlike the works of a designer, evolution by natural selection should not always lead to perfection!

Finally, it is also assumed that functional traits are specified by genes. It is genes that spread in populations, but the extent to which they do depends upon how they are expressed and thus influence the s , n , t of their bearers. In practice whether it is the genes or the traits they specify or the organisms that bear them that are considered to be the units of selection often depends on the perspective of the investigator. In principle, though, it is what endures from one generation to another that properly identifies a unit of selection and this focuses attention on the gene, in its widest sense (Dawkins, 1976), as the unit of interest — something that does have implications for functional ecology (below).

Comparative or *a posteriori* approach

'Grey is all theory, Green is life's growing tree' (Goethe) is a motto for many ecologists interested

in function (Woolhouse, 1981). What they mean is that they are more interested in investigating and documenting physiological processes than in building abstract and inevitably oversimplified models of these processes. Yet usually behind this approach is the question of why a particular physiological trait fits an organism for the ecological circumstances in which it exists. Even if the emphasis of a study is on a single physiological process in a single type of organism, it is possible to argue that the ecological approach inevitably considers processes as adaptations to environments and hence, whether consciously or not, makes value judgements about their relation to the possible success of the same traits in different environments or of different traits in the same environment. This requires an implicit, if not explicit, link between the trait and its impact on s , n and t .

A particularly powerful approach in this connection is the comparative one; comparing divergence of traits in closely related species in different environments (e.g. commonly practised by marine physiological ecologists, investigating zonation) or the divergence between traits of related organisms in similar ecological circumstance. Grime's classification of vegetation strategies is based on this latter approach (e.g. Grime, 1979) — environmental factors judged to have important and different influences on s , n , t are used to define habitats exerting different selection pressures and within these, strategies (suites of traits) are identified that do appear to converge across taxa.

A priori or hypothetico-deductive approach

Theory might be all grey, but a problem, perhaps *the* problem, with the comparative, so-called *a posteriori* (*sensu* Calow & Townsend, 1981) approaches is that no matter how detailed the physiological work might be the links that are made between function and its implications for fitness, even if explicit, are sometimes rather imprecise, so that the Panglossian problem can be acute. As is well known, correlations do not necessarily prove particular causal relationships. *A priori* models (*sensu* Calow & Townsend, 1981), on the other hand, *have* to make an explicit and precise link between a physiological function and fitness, sometimes by making auxiliary assumptions (Calow, 1984), for example that maximizing net energy returns from the process is equivalent to maximizing fitness (influential in optimal foraging

theory [Krebs & Davies, 1984]), and sometimes by making more direct assumptions about the specific links between physiological processes and s , n and t . The latter has been particularly influential in certain aspects of life-cycle theory and is discussed in detail in Sibly & Calow (1986). Many of the models in this category reduce physiology to processes of resource allocation between conflicting metabolic demands. The resources are presumed to be limiting (because of the limited nature of the resource-gathering structures and processes) and so trade-offs become important features of the models and optimality principles are used to make predictions about how organisms should be expected to function in the 'real world'. In life-cycle theory, for example, it is assumed that investment in reproduction by parents trades off with their post-reproductive survivorship. Physiological information can be used to explain this trade-off and to give insight into its specific form — this being crucial for the theory. And from information on the relative survival chances of juveniles and adults in the population, predictions can be made about the optimal investment in reproduction and hence life-cycle pattern. In principle at least, both the assumptions written into these models and the predictions that follow from them can be checked against observation. This type of approach therefore corresponds to the hypothetico-deductive model (Cousens, 1985) — make hypotheses (complex collections of which = models); use logic (suitably mathematized) to make predictions; test in controlled circumstances. Here there is also the possibility of making tests at various levels, i.e. about the underlying structure (assumptions) and overall explanatory power (predictions) of the models.

In practice, though, assumptions and predictions are often more easily made than tested. Thus with life-cycle theory it has proved surprisingly difficult to get precise information on the trade-offs (e.g. Reznick, 1985) and to carry out crucial tests on the predictions (Sibly & Calow, 1983). In the first instance, natural selection may so reduce the variance around the optimum that the trade-off itself becomes obscured (Calow & Sibly, 1983) and in the second instance, apart from the practical difficulties of obtaining life-table information for natural populations, there is the problem of dissecting ecological causes of differences between populations (e.g. due to density-dependent effects [Sutherland *et al.*, 1986]) from evolutionary ones. In these circumstances theory is not only grey in the sense discussed above but has the potential of developing a life of its own,

like a novel, internally and logically consistent, but having little bearing on the 'real world'.

Of course it is one thing to claim that the experimental programme might be hard, but it is quite another to claim that it is impossible; advocacy of *Non ultra* in biology has been a root cause, on more than one occasion, of vitalism! In the examples given above clever physiological investigation and manipulation might expose option sets and trade-off surfaces, so physiology not only benefits from but contributes to the theory (Sibly & Calow, 1986), and transplant experiments between populations in nature or to common and carefully controlled laboratory conditions can make a lot of headway in sorting out ecological from evolutionary effects (e.g. Law, 1979).

***A posteriori* and *a priori* techniques are not distinct**

Clearly, the distinction between *a posteriori* and *a priori* approaches is artificial; observations cannot be made without theory (the *tabula rasa* is a fiction) and theory makes little sense without reference to observations on the 'real world'. Yet people do continue to work hard to maintain the distinction and part of the aims of functional ecology must be to bring the two arms of the subject closer together.

Back to genes

There are also other important issues, and a particularly important one is the genotype/genome–phenotype relationship already mentioned above. Both *a posteriori* and *a priori* approaches often pay no more than lip service to genetics. Resource allocation, as with all physiological processes, is certainly modulated by enzymes so it is not unreasonable to presume that patterns of allocation are under genetic control, but a considerable amount of the variation recorded in physiological processes can be attributed to environmental rather than genetic effects *per se*. Of course, the capacity for this kind of plastic response might be genetically specified and an important issue then becomes: which conditions favour plasticity over more rigid physiological responses (Lynch & Gabriel, in press)? Additionally, it is usually assumed (again implicitly if not explicitly) that there is a direct mapping between genotype and phenotype spaces. Yet there is nothing to guarantee this and what is required here is an analysis of particular functional traits, that have known physiological effects and consequences for s , n and t , using both

classical (population/quantitative) genetics techniques and molecular ones (Watt, 1985). A very important and long-standing issue in this last context is the extent to which the extensive polymorphism exposed by electrophoretic techniques (Nei & Graur, 1984) can be linked to physiological effects with fitness consequences. Insofar as this is concerned with the relative fitness of molecules that have obvious functional effects then it falls within the compass of functional ecology.

Function, niche and group selection

Much of this essay has been concerned with the function of traits within organisms, but functional ecology will undoubtedly also be concerned with the function, or at least functioning, of organisms within communities (second sentence of Introduction). For example, 'niches' are often defined, by analogy, as the 'jobs' that organisms do within communities and this traces back to Elton (1927).

In a sense all aspects of functional ecology are concerned with this issue since they are all concerned with organisms working within environments, and part of the environment of one organism is the other organisms with which it interacts. But more is at stake here, for focusing on the 'jobs organisms do' implies that the way they contribute to the *balanced* economy of the community is an important criterion of selection. That the way organisms contribute to the economy of communities can be subject to selection is not doubted, but that this can, or indeed does, lead to a balanced economy is open to considerable debate. The possible tension between individual and 'for-the-good-of-the-group' selection has been well rehearsed (Lewontin, 1970; Maynard Smith, 1976). But the possibility that there can be certain kinds of group selection given particular (some would say very particular) requirements is now more widely countenanced (Maynard Smith, 1984). But just how balanced are the economies of communities? What is the actual frequency of occurrence of these various modes of group selection in nature and what are their implications for the functional (all senses) biology of organisms? These will certainly be questions that will be relevant for functional ecology.

Functional ecology?

To what extent, then, is there functional ecology? An answer to this question will depend in some measure on the extent to which the subject can be

identified with a paradigm and core theory (Kuhn, 1970) and the extent to which there is a common method of approach. The 'function' in functional ecology certainly covers all senses of the meaning discussed in the Introduction. It has to be about process rather than just property and if these processes are put in an ecological context then it also has to address adaptational issues. Without wishing to be too restrictive, therefore, it follows that adaptationist principles will inevitably be an important part of any 'core thinking' on the subject. Turning to methodology, functional ecology will often be concerned with physiological measurements, but insofar as it will be concerned with how these influence and can be influenced by demographic and genetical parameters it will also be concerned with population dynamics and both classical and molecular genetics. The subject is therefore broad and deep but there are surely grounds for the kind of exposure and focus being suggested here.

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